

Task 1: Formulating the World as an MDP

Week 1 Assignment — Reinforcement Learning Club

1. Scenario Description

Scenario: A student's **Daily Study Decision**. Each day the student must decide how to spend their evening hours given their current energy level and academic pressure. The goal is to maximise long-term academic performance while avoiding burnout. This is a natural sequential decision problem — each choice affects the next day's state and cumulative reward.

2. Formal MDP Definition

2.1 State Space S

Each state captures the student's current **Energy Level** and **Pending Workload**:

State ID	Energy Level	Pending Workload	Description
s0	Low	High	Tired with lots of work — dangerous zone
s1	Low	Low	Tired but caught up — needs rest
s2	Medium	High	Moderate energy, heavy backlog
s3	Medium	Low	Balanced state — sustainable
s4	High	High	Energetic but behind — grind mode
s5	High	Low	Ideal state — high energy, minimal debt

2.2 Action Space A

Action ID	Action	Description
a0	Study	Spend evening studying / doing assignments
a1	Rest	Rest, sleep early, recover energy
a2	Hangout	Social activity — fun but no work done, energy neutral

2.3 Transition Probabilities $P(s' | s, a)$

Transitions are stochastic to model real-world uncertainty (e.g., unexpected assignments, interruptions). Key rules: **Study** reduces workload (+) but drains energy (-). **Rest** restores energy (+). **Hangout** provides neither benefit. A small probability (0.1) of an unexpected workload increase always exists.

From State	Action	Next State (s')	Probability	Reasoning
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s3 (Med/Low)	Study	s2 (Med/High)	0.10	Surprise assignment appears
s3 (Med/Low)	Study	s1 (Low/Low)	0.60	Energy used, work stays low
s3 (Med/Low)	Study	s0 (Low/High)	0.10	Surprise + energy drain
s3 (Med/Low)	Study	s3 (Med/Low)	0.20	Efficient study, stay same
s3 (Med/Low)	Rest	s5 (High/Low)	0.80	Energy restored, work stays
s3 (Med/Low)	Rest	s4 (High/High)	0.20	New work appears overnight
s3 (Med/Low)	Hangout	s3 (Med/Low)	0.70	Stay same state
s3 (Med/Low)	Hangout	s2 (Med/High)	0.30	Work piles up

2.4 Reward Function $R(s, a)$

Rewards encode academic performance goals. High rewards for productive studying, penalties for burning out or ignoring deadlines.

State	Action	Reward	Justification
Any	Study	+5	Progress on academic tasks — always positive
s4/s5 High Energy	Study	+8	High-energy study is more effective
s0 (Low/High)	Study	-3	Studying when exhausted is counterproductive
Any	Rest	+2	Recovering energy is necessary maintenance
s0 (Low/*)	Rest	+6	Critical rest when very tired
Any	Hangout	-1	Opportunity cost of not studying
s0 (Low/High)	Hangout	-5	Procrastinating at worst possible time

3. Simplified 3-State MDP for Manual Value Iteration

We reduce the 6-state MDP to 3 states for tractable hand calculation:

State	Meaning
S (Study)	Student is caught up, medium energy — productive mode
R (Rest)	Student is tired — resting/recovering mode
B (Behind)	Student is behind on work — stressed mode

3.1 Simplified Transitions and Rewards

Each state has one dominant action for simplicity (deterministic transitions):

From	Action	To	Probability	Reward
S	Study	S	0.7	+5
S	Study	R	0.2	+5

S	Study	B	0.1	+5
R	Rest	S	0.8	+2
R	Rest	R	0.2	+2
B	Study	S	0.5	+3
B	Study	B	0.5	+3

3.2 Value Iteration — Two Steps ($\gamma = 0.9$)

Value Iteration update rule:

$$V^{k+1}(s) = \max_a \sum_{s'} P(s'|s,a) \cdot [R(s,a) + \gamma \cdot V^k(s')]]$$

Initialization: $V^0(S) = 0, V^0(R) = 0, V^0(B) = 0$

Iteration 1 ($k = 0 \rightarrow k = 1$):

$$\begin{aligned} V^1(S) &= \max_a \sum P(s'|S,a) \cdot [R + \gamma \cdot V^0(s')] = R(S, \text{Study}) + \\ &\gamma \cdot [0.7 \cdot V^0(S) + 0.2 \cdot V^0(R) + 0.1 \cdot V^0(B)] = 5 + 0.9 \cdot [0.7 \cdot 0 + 0.2 \cdot 0 + \\ &0.1 \cdot 0] = 5 + 0 = 5.0 \quad V^1(R) = R(R, \text{Rest}) + \gamma \cdot [0.8 \cdot V^0(S) + 0.2 \cdot V^0(R)] \\ &= 2 + 0.9 \cdot [0.8 \cdot 0 + 0.2 \cdot 0] = 2 + 0 = 2.0 \quad V^1(B) = R(B, \text{Study}) + \\ &\gamma \cdot [0.5 \cdot V^0(S) + 0.5 \cdot V^0(B)] = 3 + 0.9 \cdot [0.5 \cdot 0 + 0.5 \cdot 0] = 3 + 0 = 3.0 \end{aligned}$$

After Iteration 1: $V^1(S) = 5.0, V^1(R) = 2.0, V^1(B) = 3.0$

Iteration 2 ($k = 1 \rightarrow k = 2$):

$$\begin{aligned} V^2(S) &= 5 + 0.9 \cdot [0.7 \cdot V^1(S) + 0.2 \cdot V^1(R) + 0.1 \cdot V^1(B)] = 5 + \\ &0.9 \cdot [0.7 \cdot 5.0 + 0.2 \cdot 2.0 + 0.1 \cdot 3.0] = 5 + 0.9 \cdot [3.5 + 0.4 + 0.3] = 5 + \\ &0.9 \cdot 4.2 = 5 + 3.78 = 8.78 \quad V^2(R) = 2 + 0.9 \cdot [0.8 \cdot V^1(S) + 0.2 \cdot V^1(R)] = \\ &2 + 0.9 \cdot [0.8 \cdot 5.0 + 0.2 \cdot 2.0] = 2 + 0.9 \cdot [4.0 + 0.4] = 2 + 0.9 \cdot 4.4 = 2 + \\ &3.96 = 5.96 \quad V^2(B) = 3 + 0.9 \cdot [0.5 \cdot V^1(S) + 0.5 \cdot V^1(B)] = 3 + \\ &0.9 \cdot [0.5 \cdot 5.0 + 0.5 \cdot 3.0] = 3 + 0.9 \cdot [2.5 + 1.5] = 3 + 0.9 \cdot 4.0 = 3 + \\ &3.60 = 6.60 \end{aligned}$$

After Iteration 2: $V^2(S) = 8.78, V^2(R) = 5.96, V^2(B) = 6.60$

State	V^0 (Init)	V^1 (After Step 1)	V^2 (After Step 2)
S (Study)	0.00	5.00	8.78
R (Rest)	0.00	2.00	5.96
B (Behind)	0.00	3.00	6.60

3.3 Interpretation

After just two iterations we can already see the ordering $V^2(S) > V^2(B) > V^2(R)$. The **Study state** has the highest value because it combines a high immediate reward (+5) with a high probability (0.7) of staying in the same productive state, compounding over time. The

Behind state overtakes Rest after iteration 2 because studying from Behind leads to the high-value Study state 50% of the time. The **Rest state** has the lowest value because its immediate reward (+2) is small, even though it transitions back to the productive Study state with 0.8 probability — this benefit only materialises in later iterations. As iterations continue, all values will increase and eventually converge to the true value function V^* , encoding the long-run expected discounted reward from each state.